Dyness Balancer

Engineering algorithm and operations reference

Cerbo GX / Node-RED / Dyness RS485 / virtual BMS / DVCC

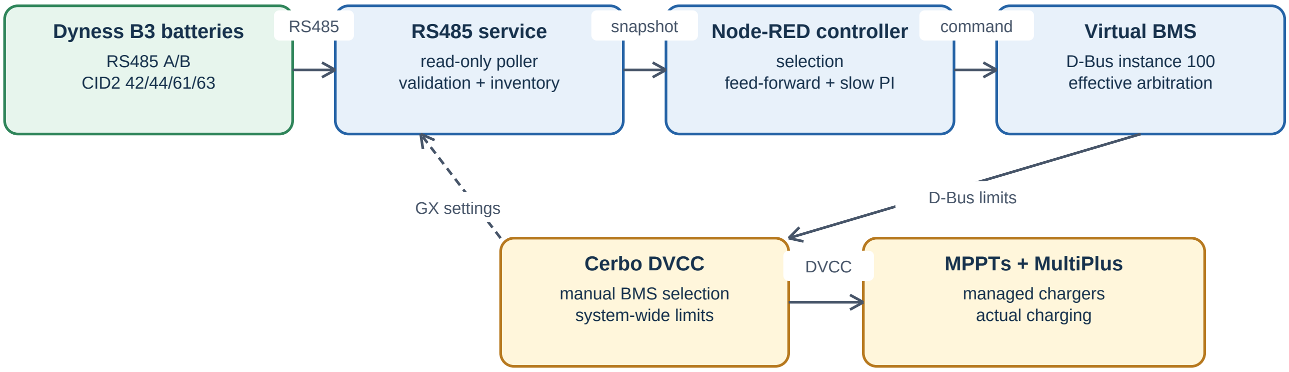
Implementation reference generated 2026-08-12. This document describes the deployed three-state controller and read-only RS485 service. It is not an authorization to enable physical charger control.

Document scope	Complete telemetry, controller, virtual-BMS, dashboard, persistence, and CSV recording behavior.
Physical authority	Cerbo BMS selection remains manual. Instance 100 applies fresh requests; CAN selection keeps them in shadow.
Control target	Hold the first selected unbalanced battery near 2.0 A by changing the aggregate charge-current request without assuming equal current sharing.
Safety fallback	Stale/incomplete telemetry or output-readback failure requests 55.0 V and 10.0 A; valid BMS and thermal limits can reduce it further.

Contents

1. Architecture and authority

The RS485 service is the only serial owner. It polls the Dyness/Pylon-compatible protocol, validates a complete snapshot, records parsed telemetry, publishes the virtual BMS, and exposes the latest snapshot to Node-RED. Node-RED calculates a versioned controller command. The service independently applies BMS, thermal, and Cerbo UI constraints before publishing virtual-BMS output.



Arrows show data/control direction. The service never writes Dyness RS485, Cerbo charge-control settings, or device-specific charger paths.

Figure 1 - System component and control data flow.

Authority hierarchy

Layer	Authority and rule
Cerbo Charge Control UI	Operator-owned maximum charge voltage and current. NORMAL requests these values; enabled limits remain final ceilings during balancing.
Dyness BMS	Authoritative CVL, CCL, DCL, charge/discharge permission, protections, and MOSFET states.
Controller	Selects an eligible battery and proposes an aggregate CCL. It never assumes equal current sharing.
Virtual BMS	Publishes final effective limits on standard D-Bus paths. It does not write charger-specific paths or GX settings.

2. RS485 telemetry and validation

All protocol requests are read-only at 115200 baud, 8N1. The polling interval is eight seconds. The controller requires complete fresh addressed CID2 0x61 telemetry for every expected battery, with a 20-second freshness limit.

CID2	Purpose	Controller authority
0x42	Addressed cell array, temperatures, signed current, battery voltage; optional capacity/lifetime tail.	Authoritative for per-battery current and logical 16-cell diagnostic vector.
0x61	Addressed integer SOC, Vmin/Vmax, packed cell locations, system voltage/current, BMS temperatures and health summary.	Sole authority for controller SOC, Vmin/Vmax, and spread.
0x63	CVL, DVL, CCL, signed DCL, permission/state bits.	Master global charge/discharge constraints and permissions.
0x44	Status1-Status5, alarms, charge/discharge/precharge MOSFETs, module power, effective charge/discharge.	Local pack eligibility/diagnostics; not a voltage-extrema source.

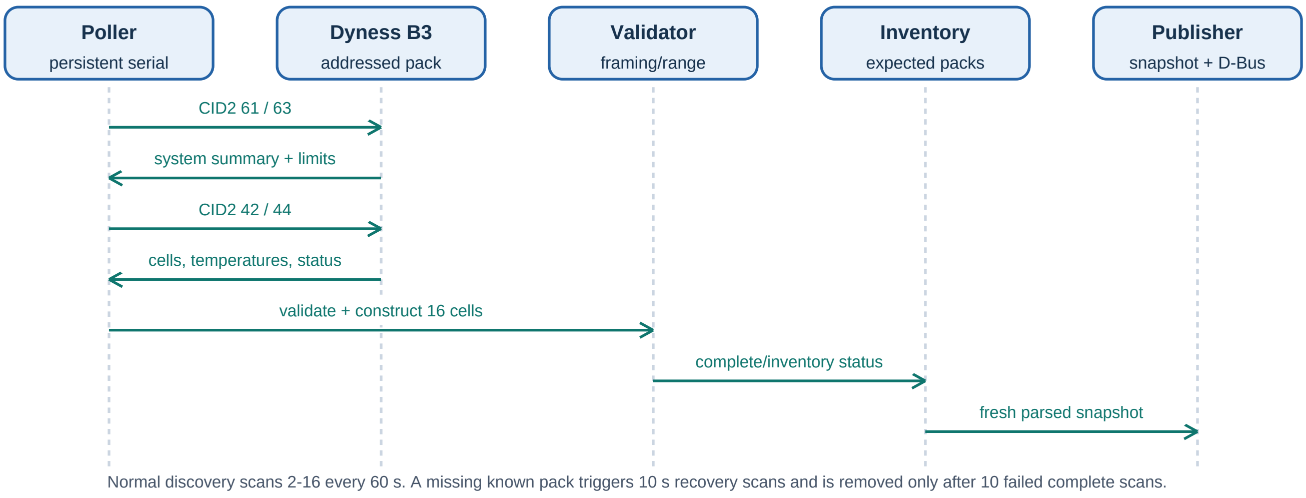


Figure 2 - RS485 polling, validation, inventory, and publication sequence.

Cell reconstruction and inventory

• Exactly 16 reported cells: retain them all. Exactly 15: derive cell 16 from that same battery's CID2 0x42 voltage minus the sum of cells 1-15. Any other count makes that battery's cell telemetry invalid.

• CID2 0x61 pack extrema are not recalculated from the 0x42 array. Invalid/unphysical CID2 0x61 extrema remain unavailable and block elevated control.

• A full address scan covers 2-16 every 60 seconds. A previously known missing battery becomes pending removal, triggers 10-second recovery scans, and is removed only after 10 failed complete scans.

3. Controller state machine

The controller has three states. It evaluates on telemetry, operator actions, and periodic ticks. Only a new telemetry timestamp advances the feed-forward/PI calculation; additional evaluations use the held request and report BALANCING_DUPLICATE_SAMPLE.

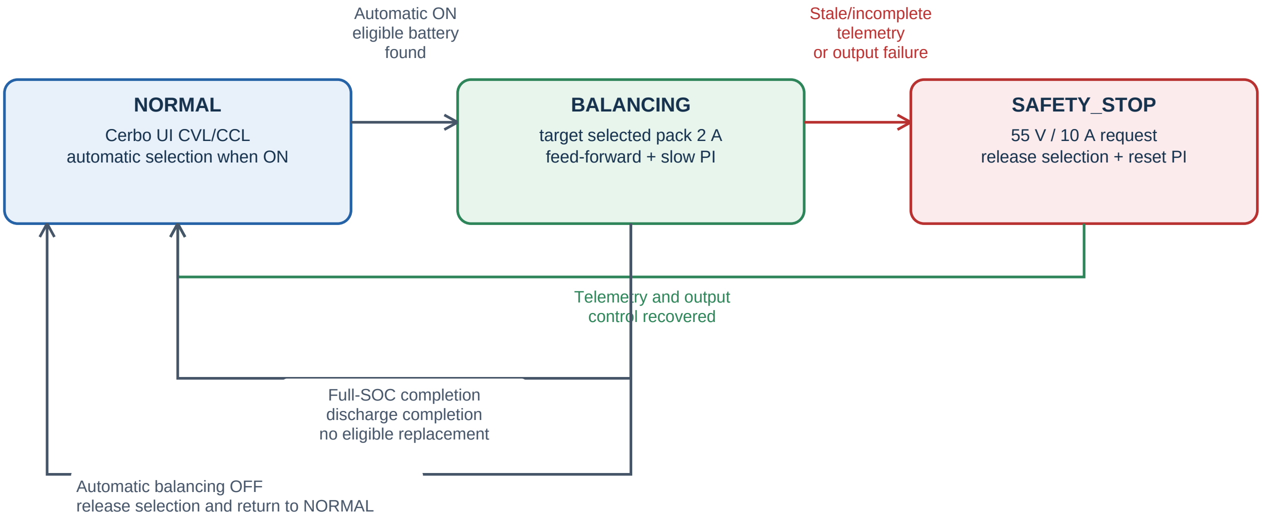


Figure 3 - Main controller state transitions.

State	Operation	Entry and exit
NORMAL	Publish normal Cerbo UI charge limits. When Automatic balancing is ON, seek the first eligible battery in ascending address order.	Remain here while Automatic balancing is OFF or no candidate exists. Enter BALANCING automatically when ON and a candidate is selected.
BALANCING	Hold the selected battery, target its measured current near 2.0 A, and adjust only aggregate CCL.	Exit to NORMAL on full-SOC completion, discharge completion, Automatic balancing OFF, or no replacement candidate. Enter SAFETY_STOP on telemetry/output failure.
SAFETY_STOP	Release selection and reset control; request conservative 55.0 V / 10.0 A while retaining charge-capable fallback semantics.	Enter on stale/incomplete telemetry, invalid configuration, or output-readback failure. Return to NORMAL only after required telemetry/control checks recover.

Eligibility and selection

• A candidate needs valid addressed CID2 0x61 data, finite current, spread strictly above 30 mV, charge MOSFET ON, and no local hard protection.

• The first qualifying address is selected and stays locked even if another battery later has a larger spread.

• If the selected battery loses its charge MOSFET or enters local protection, only it is released; the controller immediately seeks the next eligible connected battery without disabling the remaining parallel batteries.

4. Balancing current control

The controlled variable is the selected battery's measured positive current. The manipulated variable is the aggregate charge-current request. This works with unequal current sharing by estimating the selected battery's observed fraction of the positive pack current.

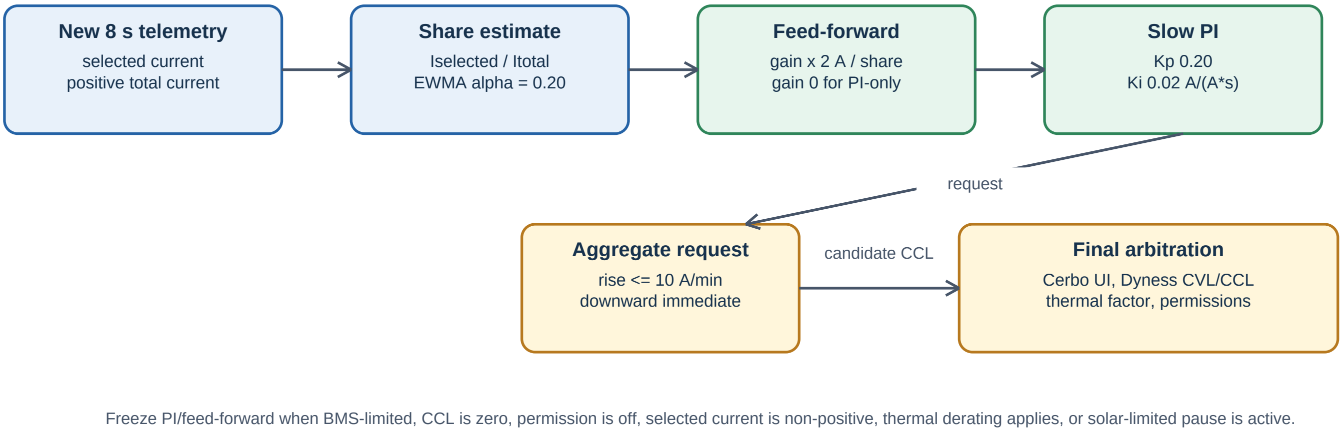


Figure 4 - Feed-forward and slow PI current-control activity.

Core calculation

```
positiveTotalCurrent = sum(max(0, current) for valid, charge-MOSFET-on, unprotected batteries)
selectedShare = selectedCurrent / positiveTotalCurrent
feedForward = targetCurrent / filteredSelectedShare
aggregateRequest = feedForward + Kp * (targetCurrent - selectedCurrent) + integral
```

Parameter	Current default	Meaning
Selected-current target	2.0 A	Desired measured current in the locked selected battery.
Spread threshold	30 mV strict	A battery is eligible only above this value.
Useful share sample	selected >= 0.25 A; total >= 0.5 A	Minimum measurement quality needed to update current-share estimate.
Feed-forward EWMA	0.20	Smooths observed sharing fraction.
Feed-forward gain	1.0	Scales feed-forward. Set 0.0 for PI-only characterization.
Kp / Ki	0.20 A/A / 0.02 A/(A*s)	Slow correction around feed-forward estimate.
Integral bound	+/- 10 A	Limits accumulated correction.
Upward slew	10 A/min	Limits increases; decreases apply immediately.
Solar detection	4 consecutive samples; 2.0 A tolerance	Approximately 32 seconds before pause/resume classification.

Solar-limited pause

When selected current is below target by more than 0.25 A and positive total current is more than 2.0 A below the held request for four samples, with no BMS/thermal/MOSFET reason, the controller remains in BALANCING but freezes share, feed-forward, P, and integral updates. It resumes after four recovery samples without resetting the held control values.

PI-only characterization

Use feed-forward gain 0.0, Kp 0.20, Ki 0.10 A/(A*s), integral bound +/-10 A, target 2.0 A, and upward slew 10 A/min. Reset control immediately before the test. The initial aggregate request remains 2 A because integration is frozen while selected current is non-positive. Stop on oscillation, BMS/thermal limitation, or unstable sharing; restore gain 1.0 and Ki 0.02 afterward.

Integral rise in A/min = Ki x current error in A x 60. At Ki 0.10, errors of 2.0, 1.5, 1.0, and 0.5 A produce 12 (slew-limited to 10), 9, 6, and 3 A/min respectively. The 10 A/min setting is a maximum output slew, not a constant PI ramp.

5. Virtual BMS / DVCC arbitration

The service accepts a versioned Node-RED command containing requested voltage, aggregate current, charge-enable intent, timestamp, and reason. Cerbo battery-monitor selection alone decides whether it is applied. The service publishes final output on standard virtual-BMS paths and retains requested-versus-effective diagnostics separately.

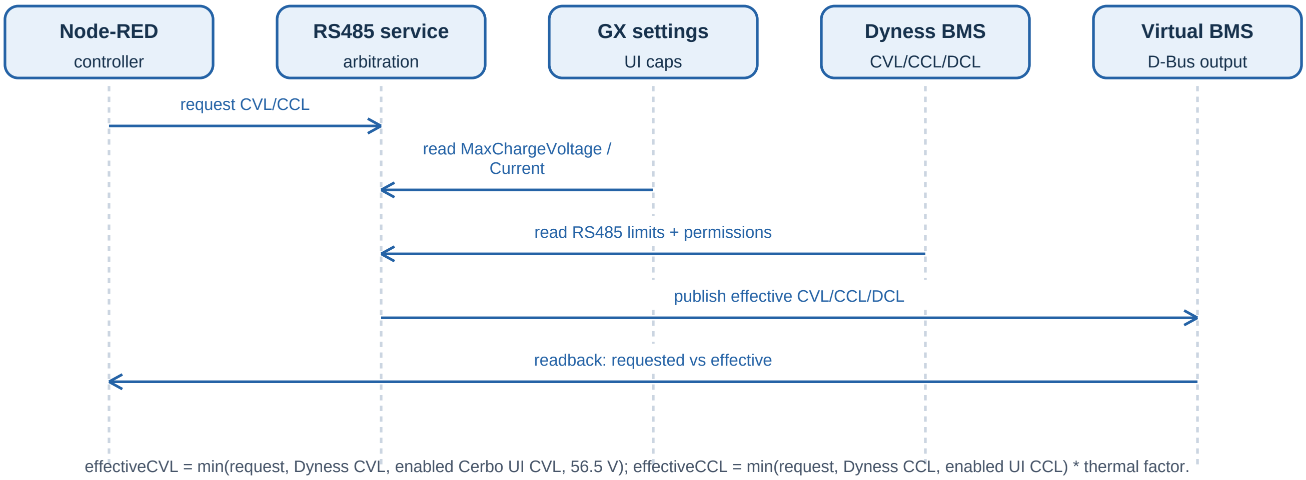


Figure 5 - Virtual-BMS arbitration sequence.

Output	Effective behavior
CVL	minimum of controller request, Dyness CVL, enabled Cerbo UI maximum charge voltage, and 56.5 V balancing ceiling.
CCL	minimum of controller aggregate request, Dyness CCL, enabled Cerbo UI maximum charge current; then multiplied by thermal factor.
DCL	Dyness DCL magnitude only while Dyness discharge permission is valid. The controller never invents an unavailable DCL.
Authority	APPLIED when instance 100 is selected; SHADOW for another valid selection; UNKNOWN when readback is unavailable.
Charge enable	Requires APPLIED authority, fresh valid context, controller charge intent, positive effective CCL, and Dyness charge permission.

Normal and safety behavior

• NORMAL follows the Cerbo values at com.victronenergy.settings /Settings/SystemSetup/MaxChargeVoltage and MaxChargeCurrent. The service only reads these settings.

• CCL = 0 and permission OFF are valid BMS constraints, not a battery disconnection. They freeze control and force effective CCL to zero.

• If telemetry is invalid, the conservative fallback is 55.0 V / 10.0 A before any valid BMS constraint can reduce it further.

6. Completion, operations, and persistence

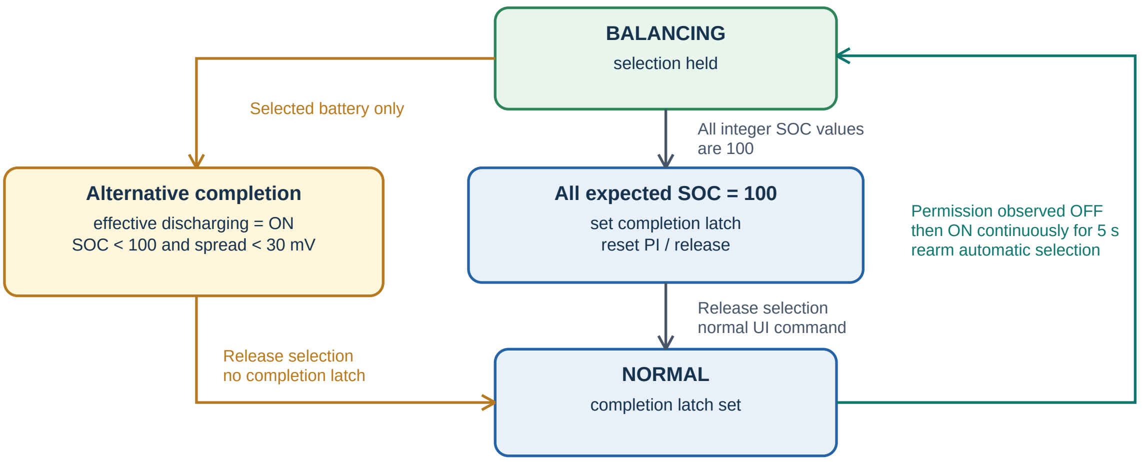


Figure 6 - Completion, latch rearm, and effective-discharge completion.

Completion rules

• Full-SOC completion: all expected batteries report integer SOC 100. The controller latches completion, resets PI, releases selection, and returns to NORMAL.

• Full-SOC latch rearm: master charge permission must be observed OFF, then remain ON for five seconds. This prevents repeated selection while batteries remain reported as full.

• Discharge completion: selected effective-discharging bit ON, selected integer SOC below 100, and selected addressed CID2 0x61 spread strictly below 30 mV. It returns to NORMAL without a full-SOC latch.

Operator controls

Control	Effect
Automatic balancing	Defaults ON after a fresh/reset state or Restore Defaults. ON selects eligible batteries automatically. OFF releases selection, resets control, and holds NORMAL using Cerbo Charge Control limits without disabling charging.
Cerbo BMS selection	The battery-monitor setting manually selects RS485 virtual BMS or normal Dyness CAN BMS and directly controls APPLIED/SHADOW authority. The balancer never changes it.
Reset control	Clears share, feed-forward, integral, and solar-pause counters without changing Automatic balancing ON/OFF.
Restore Defaults	Restores controller configuration, clears session/controller state, and turns Automatic balancing ON.
Configuration	A dirty edit buffer survives telemetry refresh. Apply waits for matching acknowledgement; Discard restores active settings.
CSV logging	Starts/stops a fixed-inventory file under /data/home/nodered/cerbo-balancer-csv/.

Persistence and retention

Controller state/configuration, command, inventory, CSV setting, events, sessions, newest snapshot, detailed parsed telemetry, and concise monthly summaries are stored below /data/home/nodered/. Detailed parsed telemetry rolls at 24 hours; the concise summary retains 30 days. Raw RS485 protocol payloads are removed before publication/persistence.

7. CSV recording format

CSV logging is intended for balancing analysis without repeating constant connection information in every row. The log file is created below `/data/home/nodered/cerbo-balancer-csv/`. Filenames must be a simple basename consisting of letters, digits, dot, underscore, or hyphen, and end in .csv.

The nominal data cadence is one valid telemetry sample per eight-second service cycle. Each data row uses the Cerbo timezone, format HH:MM:SS, plus a monotonically increasing sample_number. The combination preserves ordering even when a session exceeds 24 hours.

7.1 Constant metadata block

Metadata line	Purpose
# schema_version=12	CSV layout version.
# serial_port / # baud / # poll_interval_seconds	Constant serial and timing context.
# timestamp_format=HH:MM:SS	Clock source and display format.
# virtual_bms_service / # virtual_bms_device_instance / # dvcc_authority	Virtual BMS identity and Cerbo-selection authority context.
# status1_bits through # status5_bits	CID2 0x44 bit explanations for hexadecimal Status1-5 columns.
# initial_addresses=<addresses>	The fixed battery inventory captured at recording start.

Status1-Status5 metadata legends

Metadata key	Exact recorded bit legend
# status1_bits	bit7 pack under-voltage protection; bit6 charge-temperature protection; bit5 discharge-temperature protection; bit4 discharge over-current protection; bit3 reserved; bit2 charge over-current protection; bit1 cell under-voltage protection; bit0 over-voltage protection.
# status2_bits	bit7-bit4 reserved; bit3 module power active; bit2 discharge MOSFET ON; bit1 charge MOSFET ON; bit0 precharge MOSFET ON.
# status3_bits	bit7 effective charging; bit6 effective discharging; bit5 heater active; bit4-bit2 reserved; bit3 fully charged; bit2-bit1 reserved; bit0 buzzer active.
# status4_bits	bit7-bit0 cell voltage-check faults for cells 8-1 respectively.
# status5_bits	bit7-bit0 cell voltage-check faults for cells 16-9 respectively.

7.2 Header order and per-row groups

Order	Columns
1 - System summary	timestamp, sample_number, system_voltage_v, soc_percent, bms_temperature_c, vmin_v, vmax_v, spread_mv, battery_current_a
2 - Per battery	One complete battery_<AA>_* block for every address in initial_addresses, in ascending address order.
3 - Raw BMS limits	ccl_a, dcl_a, charge_enabled, discharge_enabled
4 - Controller request	controller_requested_voltage_v, controller_requested_current_a, controller_charge_enabled, controller_command_reason, controller_command_fresh, controller_command_age_s
5 - Final virtual BMS output	virtual_bms_effective_cvl_v, virtual_bms_effective_ccl_a, virtual_bms_effective_dcl_a, virtual_bms_charge_enabled, virtual_bms_discharge_enabled, virtual_bms_allow_to_charge, virtual_bms_allow_to_discharge, virtual_bms_thermal_factor, virtual_bms_charge_blocked_by_status, virtual_bms_discharge_blocked_by_status, virtual_bms_charge_blocked_by_controller, virtual_bms_output_valid, virtual_bms_arbitration_reason
6 - Authority and controller diagnostics	virtual_bms_authority_state, virtual_bms_controller_request_applied, controller_feed_forward_gain, controller_feed_forward_unscaled_a, controller_feed_forward_effective_a, controller_p_term_a, controller_i_term_a, controller_output_saturated, controller_output_slew_limited

7.3 Per-battery CSV block

Exact field template	Meaning
battery_<AA>_present	Battery address was present in the snapshot.

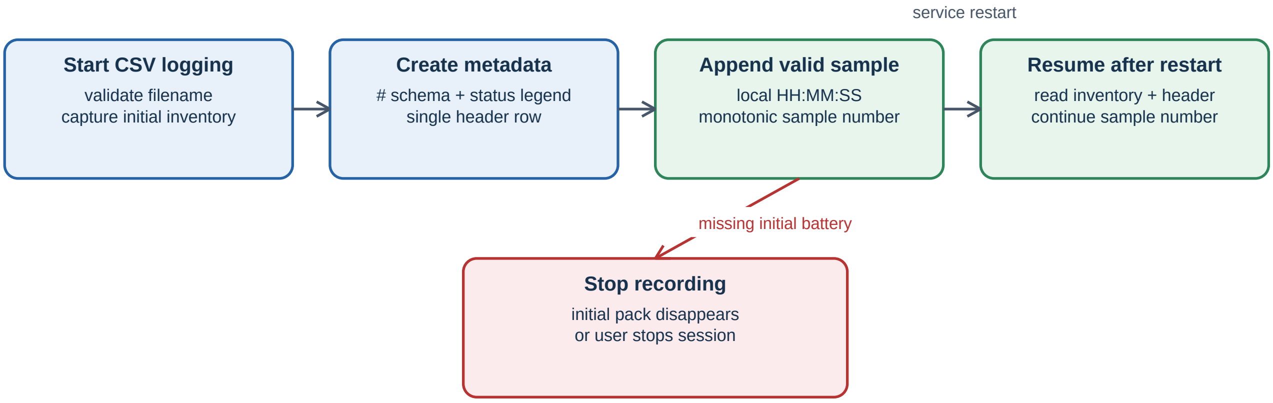
Exact field template	Meaning
battery_<AA>_valid	Validated per-battery telemetry state.
battery_<AA>_voltage_v / current_a / soc_percent	CID2 0x42 battery voltage/current and addressed CID2 0x61 integer SOC.
battery_<AA>_vmin_v / vmin_location	Addressed CID2 0x61 minimum cell voltage and decoded location.
battery_<AA>_vmax_v / vmax_location	Addressed CID2 0x61 maximum cell voltage and decoded location.
battery_<AA>_spread_mv	Addressed CID2 0x61 Vmax minus Vmin.
battery_<AA>_status1 through status5	CID2 0x44 values in fixed hexadecimal format, 0x00 through 0xFF.
battery_<AA>_cell_01_v through cell_16_v	Logical cell vector from CID2 0x42; cell 16 may be calculated from that same pack voltage.
battery_<AA>_temp_01_c through temp_05_c	First five CID2 0x42 battery temperature sensors.

Formatting and unavailable values

Value type	CSV representation
Cell voltage, Vmin, Vmax	Volts, exactly three decimal places.
Other pack voltages, currents, temperatures	Decimal values, normally two decimal places.
Spread	Whole millivolts without a decimal fraction.
Status1-Status5	Fixed uppercase hexadecimal: 0x00 through 0xFF.
Unavailable/invalid field	Empty CSV field; it is never replaced by a plausible synthetic value.

Requested versus effective

Controller-requested fields show the Node-RED proposal. virtual_bms_effective_* fields show final output after Cerbo UI limits, Dyness limits/permissions, thermal derating, freshness, authority selection, and safety rules. authority_state and controller_request_applied distinguish shadow diagnostics from output applied through selected instance 100.



Batteries added after recording starts are ignored. An existing file is appended only when its generated header matches the session schema.

Figure 7 - CSV session lifecycle.

Session invariants

• The initial active inventory fixes the header and number of per-battery blocks. Batteries added later are ignored for that session.

• If a battery from initial_addresses disappears, recording stops rather than producing misleading partial rows.

• After a service restart, the logger reads the existing metadata/header and continues the same file and sample number when its ordered columns remain a compatible subset. Columns added by a deployment begin with the next new recording file.

8. Troubleshooting and operational interpretation

Observed status	Meaning and operator response
BALANCING_DUPLICATE_SAMPLE	The controller was evaluated again before a new eight-second RS485 timestamp arrived. It intentionally holds the prior request and does not integrate PI twice. This is not a fault.
SOLAR_LIMITED_PAUSE	Cloud/available solar current is insufficient under the defined four-sample test. Selection and state remain BALANCING while control updates are frozen.
BMS_LIMITED / BMS_CCL_ZERO	Dyness BMS advertised a lower CCL or zero CCL. The controller does not counteract it; inspect BMS permission/status and temperatures.
SAFETY_STOP_TELEMETRY	Expected pack inventory or fresh addressed CID2 0x61 telemetry is missing/invalid. Verify RS485 ownership, cable, address inventory, and service health.
CAN_BMS_SELECTED_SHADOW	A fresh request is logged but normal Cerbo/BMS limits remain effective because CAN BMS is selected.
BMS_SELECTION_UNKNOWN_SHADOW	Cerbo selection readback is unavailable; the service does not apply the controller request.
FULL_SOC_COMPLETE	All expected batteries report integer SOC 100. Latch remains until master charge permission cycles OFF then ON for five seconds.
BATTERY_CHARGE_PATH_INTERRUPTED	Selected pack charge MOSFET went OFF or local protection became active. Only that pack is released; other packs continue under normal BMS/DVCC rules.

Commissioning reminder

Selecting the virtual BMS in Cerbo is manual and reversible. It immediately makes fresh valid controller requests authoritative; selecting CAN returns them to shadow. No part of this documentation authorizes an automatic handover, a charger-specific write path, disabling DVCC, or transmitting any undocumented BMS control command.

Reference implementation

The controller implementation is src/controller.js. The RS485 service, virtual BMS, arbitration, inventory, telemetry retention, and CSV logger are implemented in scripts/dyness_rs485_service.py. This PDF is generated by docs/pdf/generate_dyness_balancer_pdf.py.